

1.14.1 Properties of β function

- (i) $\beta(m, n) = \frac{\Gamma m \cdot \Gamma n}{\Gamma(m+n)}$
- (ii) $\beta(m, n) = \beta(n, m)$
- (iii) $\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$
 $\beta(n, m) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx$
- (iv) $\sin^p \theta \cdot \cos^q \theta \, d\theta = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right), (p, q > -1)$

1.15 Area between the curves

If the function $f(x) > g(x)$ for all values of x between $x=a$ and $x=b$ then

$$A = \int_a^b f(x) dx - \int_a^b g(x) dx \Rightarrow A = \int_a^b (f(x) - g(x)) dx$$

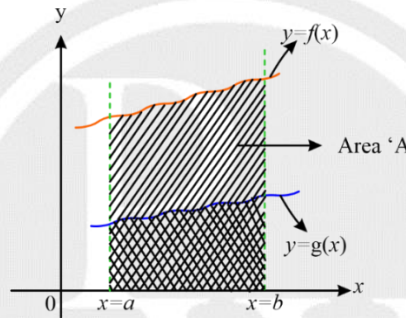


Fig. 1.11. Area under curve

Note : Area bounded by curve $r = f(\theta)$ between $\theta = \alpha$ and β is $\frac{1}{2} \int_\alpha^\beta r^2 d\theta$

1.16 Multi Variable Calculus

(a) Continuity of a function

A function $f(x, y)$ is said to be continuous at (a, b) , if $\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y) = f(a, b)$

(b) Differentiation of a two-variable function

If $f(x, y)$ is a continuous function, then the derivative of $f(x, y)$ with respect to x treating y as constant is given by

$$p = \frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

The derivative of $f(x, y)$ with respect to y treating x as constant is given by

$$q = \frac{\partial f}{\partial y} = \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k}$$

(c) Homogenous Function

A function $f(x, y)$ is said to be homogenous function of degree 'n' if $f(kx, ky) = k^n \cdot f(x, y)$.

Example:

$$f(x, y) = x^3 - 3x^2y + 3xy^2 + y^3$$

$$\Rightarrow f(kx, ky) = (kx)^3 - 3(kx)^2(ky) + 3(kx)(ky)^2 + (ky)^3$$

$$= k^3(x^3 - 3x^2y + 3xy^2 + y^3)$$

$$= k^3 \cdot f(x, y) \Rightarrow f(x, y) \text{ is a homogenous function of degree '3'.$$

(d) Euler's Theorem

If $f(x, y)$ is a homogeneous function of degree 'n' then

$$(i) \quad x \cdot \frac{\partial f}{\partial x} + y \cdot \frac{\partial f}{\partial y} = nf$$

$$(ii) \quad x^2 \cdot \frac{\partial^2 f}{\partial x^2} + 2xy \cdot \frac{\partial^2 f}{\partial x \partial y} + y^2 \cdot \frac{\partial^2 f}{\partial y^2} = n(n-1)f$$

If $f(x, y) = g(x, y) + h(x, y) + \phi(x, y)$ where $g(x, y)$, $h(x, y)$ and $\phi(x, y)$ are homogenous functions of degrees m, n and p respectively, then

$$x \cdot \frac{\partial f}{\partial x} + y \cdot \frac{\partial f}{\partial y} = m \cdot g(x, y) + n \cdot h(x, y) + p \cdot \phi(x, y)$$

$$x^2 \cdot \frac{\partial^2 f}{\partial x^2} + 2xy \cdot \frac{\partial^2 f}{\partial x \partial y} + y^2 \cdot \frac{\partial^2 f}{\partial y^2} = m(m-1) \cdot g(x, y) + n(n-1) \cdot h(x, y) + p(p-1) \cdot \phi(x, y)$$

(e) Total derivative:

$$(i) \quad \text{If } u = f(x, y) \text{ and if } x = \omega(t), y = v(t) \text{ then } \frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

$$(ii) \quad \text{If } u \text{ be a function of } x \text{ and } y, \text{ where } y \text{ is a function of } x, \text{ then } \frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dx}$$

$$(iii) \quad \text{If } u = f(x, y) \text{ and } x = f_1(t_1, t_2) \text{ and } y = f_2(t_1, t_2), \text{ then}$$

$$\frac{\partial u}{\partial t_1} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial t_1} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial t_1} \text{ and } \frac{\partial u}{\partial t_2} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial t_2} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial t_2}$$

$$(iv) \quad \text{If } x \text{ and } y \text{ are connected by an equation of the form } f(x, y) = 0, \text{ then } \frac{dy}{dx} = -\frac{\partial f / \partial x}{\partial f / \partial y}$$

(f) Concept of Maxima and Minima in Two Variables

If $f(x, y)$ is a two-variable differentiable function, then to find the maxima (or) minima.

Step-1: Calculate $p = \frac{\partial f}{\partial x}$ and $q = \frac{\partial f}{\partial y}$ and equate $p = 0, q = 0$

Let (x_0, y_0) be a stationary point.

Step-2: Calculate r, s, t where $r = \frac{\partial^2 f}{\partial x^2} \Big|_{(x_0, y_0)}$; $s = \frac{\partial^2 f}{\partial x \partial y} \Big|_{(x_0, y_0)}$; $t = \frac{\partial^2 f}{\partial y^2} \Big|_{(x_0, y_0)}$

Case (i): If $rt - s^2 > 0$ and $r > 0$, then the function $f(x, y)$ has minimum at (x_0, y_0) and the minimum value is $f(x_0, y_0)$.

Case (ii): If $rt - s^2 > 0$ and $r < 0$, then the function $f(x, y)$ has maximum at (x_0, y_0) and the maximum value is

$$f(x_0, y_0).$$

Case (iii): If $rt - s^2 < 0$; then we cannot comment on the existence of maxima and minima.

Such stationary points where $rt - s^2 = 0$ are called **saddle points**.

(g) Concept of Constraint Maxima and Minima (Method of Lagrange's unidentified multipliers).

If $f(x, y, z)$ is a continuous and differentiable function, such that the variables x, y and z are related/constrained by the equation $\phi(x, y, z) = C$ then to calculate the extreme value of $f(x, y, z)$ using Lagrange's Method of unidentified multipliers.

Step-1: Form the function $F(x, y, z) = f(x, y, z) + \lambda\{\phi(x, y, z) - C\}$, where λ is a multiplier.

Step-2: Calculate $\frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial y}$ and $\frac{\partial F}{\partial z}$ and equate them to zero

Step-3: Equate the values of λ from the above 3 equations and obtain the relation between the variables x, y and z .

Step-4: Substitute the relation between x, y and z in $\phi(x, y, z) = C$ and get the values of x, y, z . Let they be (x_0, y_0, z_0) .

Step-5: Calculate $f(x_0, y_0, z_0)$

The value $f(x_0, y_0, z_0)$ is the extreme value of $f(x, y, z)$.

(h) Multiple Integrals

If $f(x, y)$ is continuous and differentiable at every point within a region 'R' bounded by some curves is given by

$$I = \iint_R f(x, y) dx dy$$

If there is a double integral,

$$I = \int_{x=a}^{x=b} \int_{y=\phi(x)}^{y=\psi(x)} f(x, y) dy dx \quad [\text{Let } \psi(x) > \phi(x)]$$

Then I = area of the region bounded by the curves, $y = \phi(x)$; $y = \psi(x)$, $x = a$ and $x = b$ if $f(x, y) = 1$

Value of x – co-ordinate of centroid of the region bounded by $y = \phi(x)$; $y = \psi(x)$; $x = a$, $x = b$ if $f(x, y) = x$

(i) Change of Orders of Integration

$$I = \int_{x=a}^{x=b} \int_{y=\phi(x)}^{y=\psi(x)} f(x, y) dy dx \rightarrow I = \int_{y=c}^{y=d} \int_{x=g(y)}^{x=h(y)} f(x, y) dx dy$$

In change of order of Integration, the order of the Integrating variables changes and the limits as well.

Note : When limits are constants, the order of integration does not matter,

$$\int_{y=c}^{y=d} \int_{x=a}^{x=b} f(x, y) dx dy = \int_{x=a}^{x=b} \int_{y=c}^{y=d} f(x, y) dy dx$$

1.17 Jacobian of the Transformation

(i) The Jacobian of the transformation, $x = f_1(u, v)$, $y = f_2(u, v)$ is defined as,

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$$

(ii) The Jacobian of the transformation,

$x = f_1(u, v, w), y = f_2(u, v, w), z = f_3(u, v, w)$ is defined as

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix}$$

1.18 Change of Variables Formula

(i) $\iint_R f(x, y) dx dy = \iint_R f(f_1(u, v), f_2(u, v)) |J| du dv$

(ii) $\iiint_R f(x, y, z) dx dy dz = \iiint_R f(f_1(u, v, w), f_2(u, v, w), f_3(u, v, w)) |J| du dv dw$

1.19 Change of Variables

(i) Cartesian to polar co-ordinates :

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$J = r$$

$$dx dy = r dr d\theta$$

(ii) Cartesian to cylindrical polar co-ordinate :

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$J = r$$

$$dx dy dz = r dr d\theta dz$$

(iii) Cartesian to spherical polar co-ordinates :

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$J = \rho^2 \sin \phi$$

$$dx dy dz = \rho^2 \sin \phi d\rho d\phi d\theta$$

